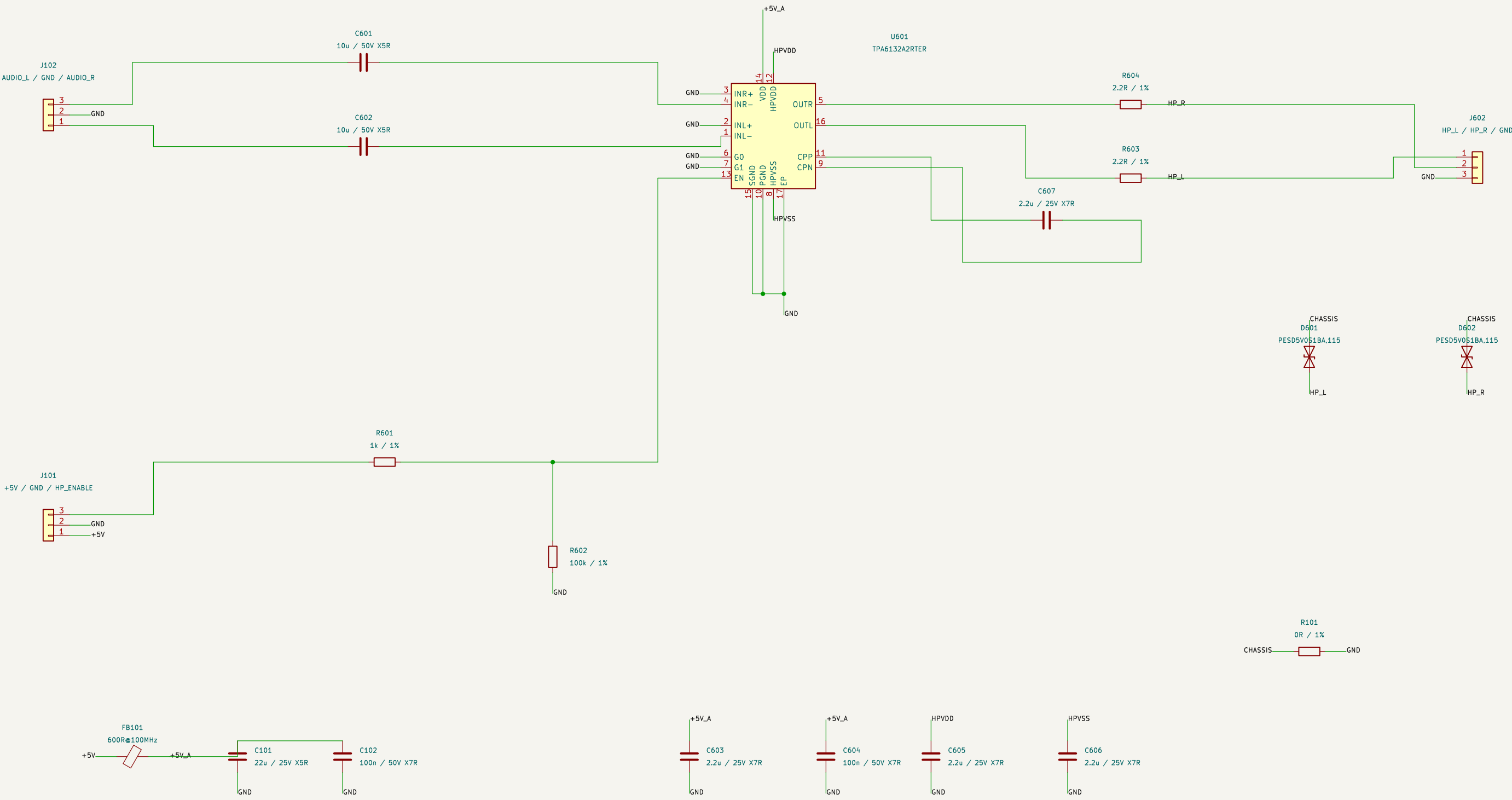


Stereo headphone amplifier / optional module

REV M1 | EXTERNAL HOST + REGULATED POWER | NO OTHER MODULE REQUIRED



Stereo headphones ≥ 32 ohm. -6 dB inverting gain (G0/G1 grounded). Inputs ≤ 1 Vrms.
No output coupling capacitors needed. HPVDD/HPVSS are local internal rails; do not connect to supply.
Enable LOW until audio is stable; ramp volume and start quietly. 5 V budget ≥ 50 mA.
J602 to TRS: 1 tip (L), 2 ring (R), 3 sleeve (GND). This optional QFN module adds assembly cost.

⏏+5V ⏏+5V_A ⏏CHASSIS ⏏GND

Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_Headphones.kicad_sch

Title: Stereo headphone amplifier / optional module

Size: A3 Date: 2026-10-02

KiCad E.D.A. 9.0.2

Rev: M1

Id: 1/1